

mCOLD Foam V.1 — Service Manual

Version 0.4 (draft) · 2026-10-05 · firmware 0.7.0-dev.8 · hardware foam V1.0.1

For technicians, firmware and QA. Deeper background: `firmware/HANDOFF.md` (state, traps, decisions), `firmware/PROTOCOL.md` (BLE/NFC/MQTT), `firmware/CHANGELOG.md` (per version).

1. Hardware overview

Block	Part	Notes
MCU	ESP32-S3-WROOM-1U-N16R8	16 MB flash, 8 MB octal PSRAM
Power path	BQ25601 charger → TPS63020 → 3V3_MAIN	SW3 switches 3V3_EN; PS/SYNC to GND (power save)
Fuel gauge	MAX17048 (0x36)	Powered from the cell; model not yet tuned to the cell
Current	INA226 (0x40), R15 10 mΩ	Battery branch; 2.5 μV = 0.25 mA per count
USB PD	HUSB238A (0x42)	PD 9 V not enabled
Temperature	MAX6675 + type-K probe	On the e-paper rail (GPIO42); T– tied to GND (rework, see 9)
Display	E-paper, mono panel for now (four-colour panel to come)	SSD1680; rotation 3
Accelerometer	LIS2DW12 (0x18)	INT1 → GPIO2, wakes the box
NFC	ST25DV04KC (0x53/0x57)	GPO → GPIO1
RTC	PCF8523 (0x68) + CR1220 on VBAT_3V	Keeps time with SW3 off
GNSS	ATGM336H on onboard antenna U3	On the SD/GNSS rail (GPIO47)
Lights	3× SK6812MINI-E front, 1× XL-4020 side	LED rail GPIO48, LOW = on
Buzzer	Magnetic, 2.7 kHz, via Q5 (GPIO6)	

Rails, all off at boot: `3V3_EPD_TK` GPIO42 high = on · `3V3_SD_GNSS` GPIO47 high = on · `3V3_LED` GPIO48 low = on.

2. Tools and connection

- PlatformIO (ESP-IDF 5.3.2), project `firmware/`, environment `mcold`.
- USB-C cable to a PC. The box appears as a USB serial port (e.g. COM7), 115200.
- Console: `pio device monitor`. DTR/RTS are held low by `platformio.ini`, so opening the monitor does **not** reset the box. Typed characters are echoed by the box; press Enter to run a command. `help` lists them.
- Only one program can hold the port: close the monitor before flashing.

3. Firmware

Step	Command
Build	<code>pio run -e mcold</code>
Flash	<code>pio run -e mcold -t upload</code>

Step	Command
Check	boot banner <code>firmware X.Y.Z</code> ; console <code>version</code>

The version is set in one place: `PROJECT_VER` in `firmware/CMakeLists.txt`. Releases are tagged `firmware/vX.Y.Z` (see `CHANGELOG.md`).

A box asleep on battery has no USB port and cannot be flashed: plug USB in (it wakes and stays awake), then flash. If `sleep_usb` is 1 the box sleeps even on USB — set it back to 0 first.

4. Console commands

Command	What it does
<code>help</code>	List of commands
<code>health</code>	Every device, its state and readings; battery, charger, rails
<code>tasks</code>	Task heartbeats, heap
<code>version</code>	Firmware version and image hash
<code>sn</code> • <code>sn set S</code>	The serial number; factory sets it, then reboot (screen, BLE, NFC, MQTT, records)
<code>accel</code> / <code>gnss</code> / <code>nfc</code>	Sensor detail
<code>led I R G B</code> • <code>ledtest [S] [N]</code> • <code>beep MS</code>	Lights and buzzer by hand
<code>config</code> • <code>config set K V</code> • <code>config reset</code>	Settings (section 5)
<code>trip</code> • <code>trip start [L H [HYST DWELL]]</code> • <code>trip stop</code> • <code>trip ack</code> • <code>trip dump [N]</code>	Trips from the bench
<code>trip csv [ID]</code>	A trip as its CSV file (<code>TRIP_<SN>_<YYMMDDhhmm>.csv</code>)
<code>log</code> • <code>logtest</code>	Trip log status; power-cut self-test (14 cases)
<code>screen</code> • <code>screen N</code> • <code>screen rot 1 3</code>	Redraw, design pages, orientation
<code>screen cal</code>	Six frames 3 px apart: count those visible per side, inset = $(6 - \text{count}) \times 3$
<code>wifi</code> / <code>sync</code>	Wi-Fi and upload status
<code>wifi add SSID PASS</code> • <code>wifi del SSID</code>	Known networks (up to 5)
<code>wifi ip SSID dhcp</code> • <code>wifi ip SSID IP GATEWAY SUBNET [DNS]</code>	A network's address: DHCP or fixed
<code>mqtt set HOST PORT [USER PASS]</code>	Broker and login (the console is the only way to set it)
<code>ota</code> • <code>ota base URL</code> • <code>ota FILE URL [force]</code>	Firmware update state, file server, install now
<code>flash</code>	Restart into download mode, to flash over USB
<code>ble [on off]</code>	BLE status; <code>on</code> advertises 60 s
<code>rpc {json}</code>	Any <code>PROTOCOL.md</code> request, authorized
<code>sleep</code> • <code>sleep clear</code>	Power manager state, wake record
<code>soc</code> • <code>soc set P</code>	Counted state of charge, capacity, anchor; set it by hand

Command	What it does
<code>reboot</code>	Restart

Bench-only (power measurement and tests):

Command	What it does
<code>amps [S]</code>	Battery current for S s: mean, min, max
<code>hiz [S] · hiz off</code>	Charger input off for S s: the box runs on its cell with USB in. Every boot switches it back on
<code>sleep test S</code>	Deep-sleep S s, timer wake only (even on USB)
<code>poweroff</code>	The battery-empty path: screen, ship mode. Unplug and replug USB to restart
<code>cpu MAX MIN LS</code>	CPU clock / light sleep until the next boot
<code>pin N 0 1 in</code>	Take a pin over as plain GPIO (until reboot)
<code>log write N · log read · log erase</code>	Bench trip records
<code>ack {json}</code>	Inject a server ACK
<code>cfgdoc {json}</code>	Inject a server settings document (docs/device-settings.md)
<code>ota rollback-test</code>	The next new image will not confirm itself

5. Settings

Stored in NVS, kept through firmware updates. `config` shows each with its range and whether it is the default.

Key	Range	Default	Meaning
<code>sample_period_s</code>	60–3600	300	Trip sample period
<code>cal_offset_c100</code>	–1000–1000	0	Temperature offset, 0.01 °C
<code>cal_gain_ppm</code>	900000–1100000	1000000	Temperature gain
<code>cal_version / cal_date</code>	—	0	Calibration record
<code>temp_adj_c100</code>	–1000–1000	0	User adjustment added after the calibration, 0.01 °C (eTEMP's tempAdj); settable from the app and the server
<code>accel_wake_ths</code>	1–63	2	Motion threshold, ×31 mg
<code>led_bright_pct</code>	1–100	20	Side light brightness cap
<code>led_front_pct</code>	1–100	2	Front lights brightness cap
<code>led_status_s</code>	60–3600	900	On battery, the alive blink this often
<code>batt_trip_mv</code>	3000–4100	3550	Below it, on battery, no new trip (BATTERY_LOW)
<code>batt_off_mv</code>	3000–3700	3400	Below it, on battery, the box switches itself off

Key	Range	Default	Meaning
<code>batt_mah</code>	300–6000	1500	Rated cell capacity, until one is learned
<code>sleep_ua</code>	0–5000	250	Current asleep, for counting charge through a sleep
<code>soc_source</code>	0–1	0	Percent shown: 0 voltage (OCV table), 1 counted
<code>tz_offset_min</code>	–720–840	420	Display time zone (records are UTC)
<code>epd_inset_t/b/l/r</code>	0–20 / 0–30	0	Panel pixels the case hides on each side (<code>screen cal</code>)
<code>buzzer_enabled</code>	0–1	1	Alarm sound
<code>sleep_en</code>	0–1	1	Deep sleep on battery
<code>idle_wake_s</code>	300–86400	3600	Longest sleep with nothing due
<code>gnss_period_s</code>	300–86400	1800	On battery in a trip: one GNSS session this often
<code>upload_period_s</code>	60–86400	300	On battery: one upload session this often
<code>sleep_usb</code>	0–1	0	Bench: behave as on battery with USB in
<code>sleep_meas</code>	0–1	0	Bench: measure each sleep's current (+0.33 mA)
<code>light_sleep</code>	0–1	0	Trial: light sleep between tasks while awake, on battery

Wi-Fi and broker credentials are separate (`wifi add` , `mqtt set`) and never appear in documents or the repository. Which keys the app and the server may set, and the settings document they use, are in `docs/device-settings.md` ; the calibration (`cal_*`), the battery limits and the bench switches are the console's only.

6. Provisioning a new box

- Flash the current release; `sn set <label SN>` , then `reboot` .
- `health` : every fitted device `ok` (`pd` , `buzzer` , `sd` show `unknown` until used).
- `logtest` : all 14 pass.
- `ledtest` and `beep 150` : check each light and the buzzer by eye and ear.
- `wifi add ...` for the site networks; `mqtt set ...` for the broker.
- `sync` : Wi-Fi connected, broker connected.
- Check the time: `health` → clock source `ntp` , `gnss` or `rtc` .

7. Power

Measured (bench, 2026-10-03)	Value
Asleep	≤ 0.25 mA (at the current monitor's floor)
Awake, radios off	~130 mA (under investigation, see 10)
Awake, BLE stack on	~155 mA
Awake, GNSS on	~173 mA
Charging from a PC port	+330 to +400 mA into the cell
Timer wake	0.13 s boot + 1.2 s awake; 3.6 s with a refresh

Behaviour on battery: deep sleep between jobs; wakes on the sample timer, motion, an NFC tap or USB power. USB power keeps the box awake.

Low battery: below `batt_trip_mv` no trip starts; below `batt_off_mv` (three readings in a row) the box records `POWER OFF` in the trip, shows BATTERY EMPTY and puts the charger in ship mode — the cell is disconnected from everything but the fuel gauge. Plugging USB in powers it back on (verified 2026-10-03).

State of charge is counted, not read from the voltage: the current monitor's readings while awake, `sleep_ua` for each sleep, 100 % when the charger reports charge done; the fuel gauge's percent only to start, or when the kept count is more than 15 % from it (charged while switched off). A run from full to the switch-off voltage teaches it the cell's real capacity. `health` shows the counted and the gauge's percent side by side. The switch-off itself is by voltage.

Wake record: `sleep` lists recent wakes (cause, boot ms, awake ms, mA, slept s). It survives a reset but not a power cut; `sleep clear` restarts it.

8. Diagnostics

health state	Meaning
<code>ok</code>	Answering
<code>degraded</code>	Worked before, failing now: something came loose
<code>failed</code>	Never worked this boot: not fitted or built wrong
<code>unknown</code>	Not tried yet

Failed devices are retried once a minute and recover on their own.

Trip records (`trip dump N`): SAMPLE (temperature, motion, GNSS, battery, alarms), EVENT (RESUMED, MOTION, PROBE FAULT/OK, ALARM, ALARM CLEAR, ALARM ACK, TIME SET, LOSS, POWER OFF), START, STOP.

9. Known hardware rework (every board of this BOM)

Item	Do
GNSS RF path	Leave off C_M1, C_M2 and R_SET_IPEX (onboard antenna). For the u.FL instead: off R_SET_PCB, C_M1, C_M2; keep R_SET_IPEX
RTC backup	CR1220 to VBAT_3V (pad hand-soldered on V1.0.1)
Thermocouple	MAX6675 T- to GND, or an open probe is not detected
Buzzer	Its pulse sets off the accelerometer; firmware ignores motion during a beep

10. Troubleshooting

Symptom	Check
No USB port	Box asleep on battery: plug in USB. Still none: SW3, then the battery
pio upload fails, port busy	Close the serial monitor
Box dark after poweroff or an empty battery	Unplug and replug USB. If not: disconnect and reconnect the battery
Temperature --	Probe plugged in? <code>health</code> → thermo

Symptom	Check
Clock -----	No source yet: Wi-Fi (NTP), sky (GNSS), or the app; check the CR1220
No GNSS fix	Indoors is normal; sessions give up after 60 s with nothing heard
Records never leave the box	Server not sending ACKs (<code>sync</code> shows pending)
Battery % over 100 / "no battery"	No cell connected: the gauge sees the charger's output
Motion events with nobody near	Under investigation; buzzer and rail switching are already blanked
Awake current ~130 mA	Under investigation (bench floor ~40 mA)

Not yet included

- Battery: cell datasheet, charge current and limits, fuel-gauge model, runtime
- Root cause of the ~130 mA awake current
- Charger / PD policy, HUSB238A register map, PD 9 V
- USB drive with CSV (phases F2-F4), SD archive (deferred)
- Dock (6 boxes), factory provisioning tool, production logging, secure boot / flash encryption (P8)
- Temperature calibration procedure and accuracy
- Four-colour display, shock alarm, door sensor
- Disassembly, part replacement, schematics references by page